

GENERATIVE AI IN VISUAL DESIGN: Investigating the Role of Human Involvement in Visual Communication

1. Project Status

During Weeks 5 and 6, the project moved from methodology development into the creation of the experimental materials. Progress Report 1 established the three design conditions, the general structure of the standardized briefs, the visual design principles that will be evaluated, and the plan for participant testing. The next stage identified in that report was to finalize the briefs, create the experimental flyers, and document the different design processes. REPORT 1 senior seminar

The main work during this reporting period included finalizing the three promotional contexts, developing standardized briefs, refining what information should remain controlled across conditions, developing instructions for each design condition, creating the three AI-generated flyers, and beginning recruitment for the human-designed and AI-assisted conditions.

The three promotional contexts are now:

- Event promotion: **Culture Fest**
- Health/public awareness: **Know Your Numbers**
- Career/recruitment: **FutureTech Fellowship**

All three take place within the fictional context of Riverton University.

2. Development of the Standardized Briefs

Progress Report 1 proposed using three standardized campus-oriented event topics. During this reporting period, I reconsidered using three events because I wanted the study to examine visual communication across different promotional purposes rather than repeatedly testing the same type of communication.

I therefore selected three different promotional contexts. Culture Fest focuses on attracting attention, communicating event information, and encouraging attendance. Know Your Numbers focuses on communicating health information clearly and encouraging students to take an action. FutureTech Fellowship focuses on communicating an opportunity, its benefits and eligibility requirements, and encouraging students to apply.

This variation allows the study to examine the three design processes under different communication demands while keeping the overall format consistent as promotional flyers.

Each brief contains four main components: a scenario, target audience, required factual information, and communication objective. The briefs intentionally do not specify colors, typography, layout, imagery, information hierarchy, or which information should receive the most visual emphasis. These decisions are left to the designer or AI system.

3. Refining Standardization and Design Autonomy

One of the most important methodological decisions during this reporting period involved reconsidering what it means for the information to remain “consistent” across conditions.

Progress Report 1 established that each version should use the same required information so that the study compares differences in the design process rather than differences in the information being communicated. However, I identified a potential problem with interpreting consistency as requiring identical wording.

Part of human design involves making communication decisions. A designer may recognize that a sentence is unnecessarily long, that certain information needs to be simplified, or that a call to action can be communicated more effectively using different wording. Preventing these decisions would remove part of the human judgment that this research is intended to investigate.

For this reason, I refined the control so that **factual content remains standardized, while presentation does not**. Every condition receives the same facts and communication objective. However, wording, information hierarchy, organization, visual emphasis, and presentation may vary. Designers may rephrase information for clarity as long as its factual meaning is not changed.

This allows the experiment to maintain comparable source information without controlling away the communication decisions that may distinguish human-designed, AI-assisted, and AI-generated work.

4. Fictional University Context

Another decision was to use a fictional institution, **Riverton University**, for all three briefs rather than an existing university.

Using a real institution could introduce existing brand associations, including familiar colors, logos, typography, or expectations about how university materials should look. The fictional university provides enough context for the flyers to feel realistic without requiring designers to follow an established visual identity.

No university brand colors or design system are provided. Therefore, decisions involving color, typography, imagery, and visual style remain part of the design process being examined.

5. Design Condition Instructions

Progress Report 1 established operational definitions for the human-designed, AI-assisted, and AI-generated conditions. During Weeks 5 and 6, I translated these definitions into instructions that can be consistently provided during flyer production.

The introductory instructions remain as similar as possible across conditions. Each condition receives the same scenario, audience, required information, and communication objective and is given freedom to determine wording, hierarchy, layout, typography, colors, imagery, and visual presentation.

For the **human-designed condition**, generative AI cannot be used during any stage of the process. This includes AI assistance with wording, layout, visual concepts, imagery, color palettes, or typography.

For the **AI-assisted condition**, the human remains the primary designer but is required to use generative AI during the creative process. AI may contribute to areas such as wording, layout exploration, visual concepts, colors, typography, or imagery. The designer can accept, reject, combine, or modify AI suggestions and remains responsible for the final design. AI-assisted designers will also document how AI was used.

For the **AI-generated condition**, the same standardized brief is provided directly to the generative AI system. The AI is allowed to determine wording, hierarchy, layout, typography, colors, imagery, and overall presentation. Additional human aesthetic direction is not provided because doing so would introduce human design decisions into the AI-generated condition.

6. AI-Generated Flyer Production

The three flyers for the AI-generated condition were completed during this reporting period. One flyer was generated for each standardized brief: Culture Fest, Know Your Numbers, and FutureTech Fellowship.

A standardized instruction was used so that the AI received the communication problem without being given specific visual solutions. The AI was instructed to create a promotional flyer from the scenario, target audience, required information, and communication objective while keeping factual information accurate and avoiding the invention of additional factual details.

This approach was important because directing the AI to emphasize a particular fact, use a particular layout, or choose a specific visual style would introduce human design judgment into a condition intended to represent AI as the primary designer.

The completion of these three flyers marks the first completed experimental condition of the study.

7. Designer Recruitment

Recruitment has also begun for the human-designed and AI-assisted conditions.

The initial recruitment message was intentionally kept general. Potential designers are told that the project involves creating three promotional flyers from supplied briefs, but the message does not explain the specific visual principles or communication measures that will later be used to evaluate the designs.

This was done to reduce the possibility of designers deliberately optimizing their work for the study's evaluation criteria. Designers should understand the design task and communication objective, but they do not need to know the exact measures that participants will later use to evaluate their work.

The current baseline plan remains to have at least one independent designer complete all three briefs within each human-involved condition. However, if recruitment allows, I would like to include multiple designers. Having three designers complete all three briefs would provide greater representation of variation in human design decisions than relying on a single designer.

8. Evidence of Progress

During Weeks 5 and 6, I:

- Finalized the three promotional contexts.
- Developed standardized briefs for all three contexts.
- Changed the experimental materials from three similar event topics to three different promotional communication contexts.
- Refined standardization so that factual information remains controlled without requiring identical wording or presentation.
- Selected a fictional university context to reduce the influence of existing branding.
- Developed consistent instructions for the human-designed, AI-assisted, and AI-generated conditions.
- Established that designers will know the communication objective but not the specific evaluation criteria.
- Completed all three AI-generated experimental flyers.
- Began recruiting designers for the human-designed and AI-assisted conditions.
- Developed a plan to document AI use and design decisions during the AI-assisted process.

This represents a transition from methodology planning into actual experimental production. Progress Report 1 described the primary evidence at that stage as the development of a more specific and testable methodology and identified flyer creation as the next source of tangible evidence. The completed briefs and AI-generated flyers now provide that next stage of evidence.

9. Challenges and Methodological Decisions

The main challenge during this reporting period was balancing **experimental consistency with design autonomy**.

Initially, I considered keeping all information identical across the flyers, including wording. However, this raised a methodological concern because rewriting, simplifying, organizing, and emphasizing information are themselves design decisions. Requiring identical wording could make the comparison more controlled, but it could also remove an important area where human judgment may contribute to communication effectiveness.

I therefore decided to standardize the facts rather than the exact presentation of those facts. This creates some additional variation between the outputs, but that variation is intentional because the study is investigating how different design processes communicate the same underlying information.

Another challenge involved determining how much information designers should receive about the study itself. The visual principles established in Progress Report 1 include hierarchy, typography, contrast, alignment, spacing, composition, and balance. Providing this evaluation framework directly to designers could influence how they approach the task. I therefore decided that designers will receive the communication objective but will not be told the exact criteria that participants will later evaluate.

A final consideration is the number of human designers. One designer completing all three briefs provides consistency and keeps the study manageable, but it limits how representative the human-designed condition can be. If enough designers are available, I plan to expand the human-involved conditions while keeping each designer responsible for all three briefs so that designer differences are not confused with differences between promotional contexts.

10. Next Steps

The immediate next step is to complete the **human-designed and AI-assisted flyers**. Designers who agree to participate will receive the appropriate condition instructions followed by the same three standardized briefs used for the AI-generated condition.

During production, I will collect information about production time and, where applicable, AI prompts, AI contributions, human modifications, and design decisions. This continues the documentation plan established in Progress Report 1, which identified production time, human intervention, prompts, and design decisions as important data for the study. REPORT 1 senior seminar

Once the experimental flyer set is complete, the next major stage will be developing and piloting the participant evaluation. The evaluation will measure both perceived design quality and communication effectiveness. Progress Report 1 proposed participant ratings for visual appeal, readability, organization, professionalism, clarity, and overall effectiveness, along with comprehension questions. REPORT 1 senior seminar

The participant evaluation will be refined around the final flyer set before data collection begins.

11. Conclusion

Weeks 5 and 6 moved the project from defining the experiment to actually producing its materials. The three standardized briefs are now complete, the AI-generated flyers have been created, condition-specific instructions have been established, and recruitment has begun for the human-designed and AI-assisted conditions.

The most important methodological development was recognizing that standardization should preserve **the same factual information without eliminating design judgment**. This allows each condition to respond to the same communication problem while still making its own decisions about how that information should be communicated.

The next stage will complete the remaining experimental flyers and prepare the participant evaluation so that the study can begin collecting data on both visual quality and communication effectiveness.

